

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>
Key facts about fetal and early postnatal brain development support that the fetal cortex is the last major system to develop .

- In utero : The brain progresses from ancestral to advanced structures . In evolutionary terms , the earliest parts to appear are the brain stem and cerebellum ; the oldest region to be born is the thalamus . From that point onward , development slows .
- Postnatally : The frontal cortex and ventral portions of the cortex (including primary somatosensory , motor , and association regions) have organized activity and connections in the second trimester of pregnancy and first year of life . By contrast , the dorsal cortex (primary visual and auditory regions) becomes functionally specialized later , in parallel with the maturation of receptive pathways in the thalamus . The cortical ‘stripes’ containing layer 4 are especially slow to organize compared to other layers , reflecting their key role in sensory perception .
- Pathways : Primary visual system tracts (e.g . , visual cortex to superior colliculus via the pulvinar) undergo a protracted maturation trajectory and are prominent indicators of cortical functional development . At birth , the visual system is much more mature than the frontal cortex , even though the cortex is more recently evolved in phylogeny .

Therefore , the part of the brain that remains least developed at birth is the cortex (specifically the dorsal cortex / visual system) , not the thalamus or basal ganglia , both of which are organized earlier .

Correct answer : G . cortex

</think>
<answer>
G . cortex
</answer>